

Mahdi Mokhtarzadeh

Heerweg-Zuid 102D, Ghent 9052, Belgium

(+32) 470 53 27 04; <https://users.ugent.be/~mmokhtar/>; mahdi.mokhtarzadeh@ugent.be**PROFILE:**

Researcher assistant with a strong foundation in operations research and optimization, combined with experience implementing optimization algorithms in C++, MATLAB, and Python. Author of 15+ peer-reviewed papers on supply chain, scheduling, and quality management. Passionate about using optimization techniques to solve real-world supply-chain and production problems.

RESEARCH INTERESTS:

- Operations Research and Management
- Quantum Computing in Optimization and Machine Learning
- Probabilistic Reasoning and Causal Inference
- Production Scheduling, *Seru* Production Systems, and Line Balancing
- Quality Management, Failure Analysis, and Reliability Engineering
- Industry 4.0 and 5.0 and Smart Manufacturing
- Supply chain analysis and management

COMPUTER SKILLS:

- *Programming Languages*: MATLAB, Python, C, and C++
- *Optimization Software*: IBM ILOG CPLEX, GAMS, Gurobi, D-Wave Quantum Annealing, and IBM Qiskit
- *Other Tools*: MS Excel, Minitab, and MS Project

LANGUAGE SKILLS:

- Persian: *Native*
- English: *Professional* (TOEFL: 102/120)
- Dutch: *Intermediate* (B1)

EDUCATION:

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| Apr. 2022 – Dec. 2026 | Ph.D., Industrial Engineering, Ghent University, Ghent, Belgium,
Thesis Title: Probabilistic Reasoning for Quality Management in Industry 4.0, Supervisor: Prof. Sidharta Gautama |
| Sep. 2017 - Jul. 2020 | M.Sc., Industrial Engineering, University of Tehran, Tehran, Iran, GPA: 19.21/20,
Thesis Title: Production Planning of Mixed Model Assembly Lines by Considering Human Factors and Customer Satisfaction, Supervisor: Prof. Masoud Rabbani |
| Sep. 2011 - Jan. 2015 | B.Sc., Industrial Engineering, University of Hormozgan, Bandar Abbas, Iran, GPA: 16.24/20
Thesis Title: Statistical Analysis of Refractory Brick Life in Cement Factory Furnace to Determine Optimal Brick Layout, Supervisor: Dr. Yousef Rabbani |

SELECTED COURSES:

Probabilistic Graphical Models, Bayesian Statistics, Causal Machine Learning, Queuing Systems, Game Theory, Operations Research, Discrete Optimization, Advanced Engineering Economics

PROFESSIONAL EXPERIENCE:

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| Apr. 2022 – Present | Research Assistant, Department of Industrial Systems Engineering and Product Design, Ghent University, Ghent, Belgium; in collaboration with Flanders Make, Supervisor: Prof. Sidharta Gautama
Project: Assembly Quality Management Using System Intelligence |
| Sep. 2026 – Dec. 2026 | Research Visit, GEMTEX Textile Research Laboratory, ENSAIT, Lille University, Lille, France
Project: Towards Human-Centered Decision Support Engineering for Intelligent Textile Manufacturing, Supervisor: Prof. Kim-Phuc Tran |
| Jan. 2020 – Feb. 2021 | Researcher Assistant, Operations Management, University of Washington, Supervisor: Prof. Thomas Schmitt
Project: Adapting Supply Chain Systems to Replenish Critical Shortages |
| Jul. 2018 – Feb. 2022 | Freelance Operations Research / Optimization Developer, Tehran, Iran
Projects: Developed mathematical models and (meta)heuristics for supply chain and production problems, including location-allocation, VRP, and scheduling, using C++, MATLAB, Python and optimization solvers |

- Feb. 2019 – Research Assistant, University of Tehran, Tehran, Iran, Supervisor: Prof. S. Ali Torabi
 Dec. 2020 Project: Multi-Criteria Decision Making for Process Safety Performance Management in Petrochemical Industries
 Oct. 2018 – Teaching Assistant, University of Tehran, Tehran, Iran, Main Lecturer: Prof. Masoud Rabbani
 Jul. 2021 Advanced Production Planning, Advanced Engineering Economics
 Jun. 2015 – Project Control / Planning Engineer, Railway Construction, Shiraz, Iran
 Feb. 2017 Project schedules, reports, and engineering tasks for a railway infrastructure project based on PMBOK standard
 Jul. 2014 – Internship, Cement Factory, Darab, Iran,
 Sep. 2014 Project: Implementation of 5S Methodology in the Machinery Workshop

SELECTED PUBLICATIONS:

ORCID: 0000-0002-0348-6718, Full list: [Google Scholar](#)

Peer-Reviewed:

1. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Gautama S. (2026). pFMEA-based Decision-Support Systems for Quality Management: Case Study. *29th International Conference on Information Fusion*.
2. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Gautama S. (2025). A Framework for Integrated Data-Supported Failure and Root Cause Analysis. *Reliability Engineering & System Safety*, 269, 112099.
3. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Semanjski I., Gautama S. (2025). Hybrid Intelligence Failure Analysis for Industry 4.0: A Literature Review and Future Prospective. *Journal of Intelligent Manufacturing*, 36, 2309–2334.
4. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Zeren Z., Van Noten J., Gautama S. (2024). Pure Data-Driven Machine Learning Challenges for pFMEA: A Case Study, *IFAC-PapersOnLine*, 58 (19), 658-663.
5. Farsi A., **Mokhtarzadeh M.**, Rabbani M., Manavizadeh N., Ghasempour Anaraki M. (2024). Using Parallel Metaheuristics to Solve a Parallel U-shaped Robotic Mixed-model Assembly Line Balancing and Sequencing Problem. *Soft Computing*, 28, 12603–12621.
6. Steck K., **Mokhtarzadeh M.** (2022). Balancing Collaborative Human-Robot Assembly Lines to Optimize Cycle Time and Ergonomic Risk. *International Journal of Production Research*, 60 (1), 25-47.
7. **Mokhtarzadeh M.**, Tavakkoli-Moghaddam R., Triki C., Rahimi Y. (2021). A Hybrid of Clustering and Meta-Heuristic Algorithms to Solve a *p*-Mobile Hub Location–Allocation Problem with the Depreciation Cost of Hub Facilities. *Engineering Applications of Artificial Intelligence*, 98, 104121.
8. Triki C., Mahdavi Amiri M., Tavakkoli-Moghaddam R., **Mokhtarzadeh M.**, Ghezavati V. (2021). A Combinatorial Auction-Based Approach for the Ridesharing in a Student Transportation System. *Networks*, 78, 229–247.
9. **Mokhtarzadeh M.**, Rabbani M., Manavizadeh N. (2021). A Novel Two-Stage Framework for Reducing Ergonomic Risks of a Mixed-Model Parallel U-Shaped Assembly-Line. *Applied Mathematical Modelling*, 93, 597-617.
10. **Mokhtarzadeh M.**, Tavakkoli-Moghaddam R., Vahedi-Nouri B., Farsi A. (2020). Scheduling of Human-Robot Collaboration in Assembly of Printed Circuit Boards: A Constraint Programming Approach. *International Journal of Computer Integrated Manufacturing*, 33(5), 460-473.
11. Taleizadeh A., **Mokhtarzadeh M.** (2020). Pricing and Two-Dimensional Warranty Policy of Multi-Products with Online and Offline Channels Using a Value-At-Risk Approach. *Computers & Industrial Engineering*, 148, 106674.
12. Farsi A., Torabi SA., **Mokhtarzadeh M.** (2023). Integrated Surgery Operations Scheduling by Constraint Programming and Meta-heuristics. *International Journal of Management Science and Engineering Management*, 4(18) 292-304.
13. Fatemi-Anaraki S., **Mokhtarzadeh M.**, Rabbani, M., Abdolhamidi D. (2022). A Hybrid of K-Means and Genetic Algorithm to Solve a Bi-Objective Green Delivery and Pick-Up Problem. *Journal of Industrial and Production Engineering*, 39(2) 146-157.
14. Rabbani M., **Mokhtarzadeh M.**, Manavizadeh N., Farsi A. (2021). Solving a Bi-Objective Mixed-Model Assembly-Line Sequencing Using Metaheuristic Algorithms Considering Ergo-nomic Factors, Customer Behavior, and Periodic Maintenance. *OPSEARCH*, 58(3), 513-539.
15. Rabbani M., **Mokhtarzadeh M.**, Manavizadeh N. (2021). A Constraint Programming Approach and a Hybrid of Genetic and K-Means Algorithms to Solve the *p*-Hub Location-Allocation Problems. *International Journal of Management Science and Engineering Management*, 16(2), 123-133.
16. Rabbani M., Abdolhamidi D., **Mokhtarzadeh M.**, Fatemi-Anaraki, S. (2021). Solving a bi-objective medicine distribution problem considering delivery to waste center using a hybrid clustering, mathematical modeling and NSGA-II approach. *Journal of Industrial and Systems Engineering*, 13(2), 245-263.

17. **Mokhtarzadeh M.**, Jolai F., Nikoubin A. Afshar-Bakeshloo M. (2019). Determining Maintenance Opportunity Window (MOW) in Job-Shop Systems by Considering Manpower of Maintenance. *Journal of Quality Engineering and Production Optimization*, 4(1), 41-54.
18. Rabbani M., **Mokhtarzadeh M.**, Farrokhi-Asl H. (2018). A New Mathematical Model for Designing a Municipal Solid Waste System Considering Environmentally Issues. *International Journal of Supply and Operations Management*, 5(3), 234-255.
19. Farsi A., **Mokhtarzadeh M.**, Afshar-Bakeshloo M. (2019). Berths Allocation Problem by Considering Probability of Disturbance in the Berths. *12th International Conference of the Iranian Operations Research Society, in Persian*.

Works in Progress:

1. Schmitt T., **Mokhtarzadeh M.**, Stecke K., Kovacs K. (2026). Digital and Quantum Steps towards Profit in Job Shop Scheduling.
2. **Mokhtarzadeh M.**, Rodríguez-Echeverría J., Gautama S. (2026). A Context-Aware Human-in-the-Loop Decision Support System for Failure Analysis in Smart Manufacturing.
3. **Mokhtarzadeh M.**, Stecke K. (2026). Competitive Joint Network Design and Pricing Problem for On-demand Drone Delivery.

PRESENTED at CONFERENCES and SEMINARS:

2026, 29th International Conference on Information Fusion (FUSION 2026)
 2025, Flanders Make Symposium 2025 - Where Science Meets Solutions
 2025, 34th European Conference on Operational Research (EURO 2025)
 2024, 18th IFAC Symposium on Information Control Problems in Manufacturing (INCOM 2024)

HONORS:

2022, Awarded Research Assistant Scholarship, Ghent University
 2020, Admitted to Ph.D. Program in Operations Management, University of Texas at Dallas
 2018-2019, Ranked First in the Class with a GPA of 19.21/20, University of Tehran
 2017, Top 0.5% (54/10,000) in Iran National University Entrance Exam for a M.Sc. in Industrial Engineering

HOBBIES and INTERESTS:

Traveling, Hiking, Swimming, Table Tennis, History and Philosophy, and Literature (Poetry and Fiction)

REFERENCES:

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- Prof. Kathryn Stecke, *Naveen Jindal School of Management, University of Texas at Dallas, Dallas, TX*, KStecke@utdallas.edu
 - Prof. Thomas Schmitt, *Foster School of Business, University of Washington, Seattle, WA*, glennsch@u.washington.edu
 - Prof. Sidharta Gautama, *Department of Industrial Systems Engineering and Product Design, Ghent University, Ghent, Belgium*, Sidharta.Gautama@UGent.be
 - Prof. Masoud Rabbani, *Department of Industrial Engineering, University of Tehran, Tehran, Iran*, mrabani@ut.ac.ir
 - Prof. Reza Tavakkoli-Moghaddam, *Department of Industrial Engineering, University of Tehran, Tehran, Iran*, tavakoli@ut.ac.ir